The many times of action observation on corticospinal excitability: Coupling online facilitation with after-effects.

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Does observation leave a trace?

Online motor resonance is well established. Does it leave a measurable corticospinal trace **after observation**?

We tested whether individual Action Obsevation Responsiveness predicts subsequent resting excitability.

Experimental design

32 healthy right-handed adults
Within-subject design

BASELINE
Rest

⬇

ACTION OBSERVATION
VR · TMS during observation

4 conditions

Ego 2D	**Ego 3D**
Allo 2D	**Allo 3D**

⬇

POST-SESSION
Rest

TMS: left M1 EMG: ECR / FDI

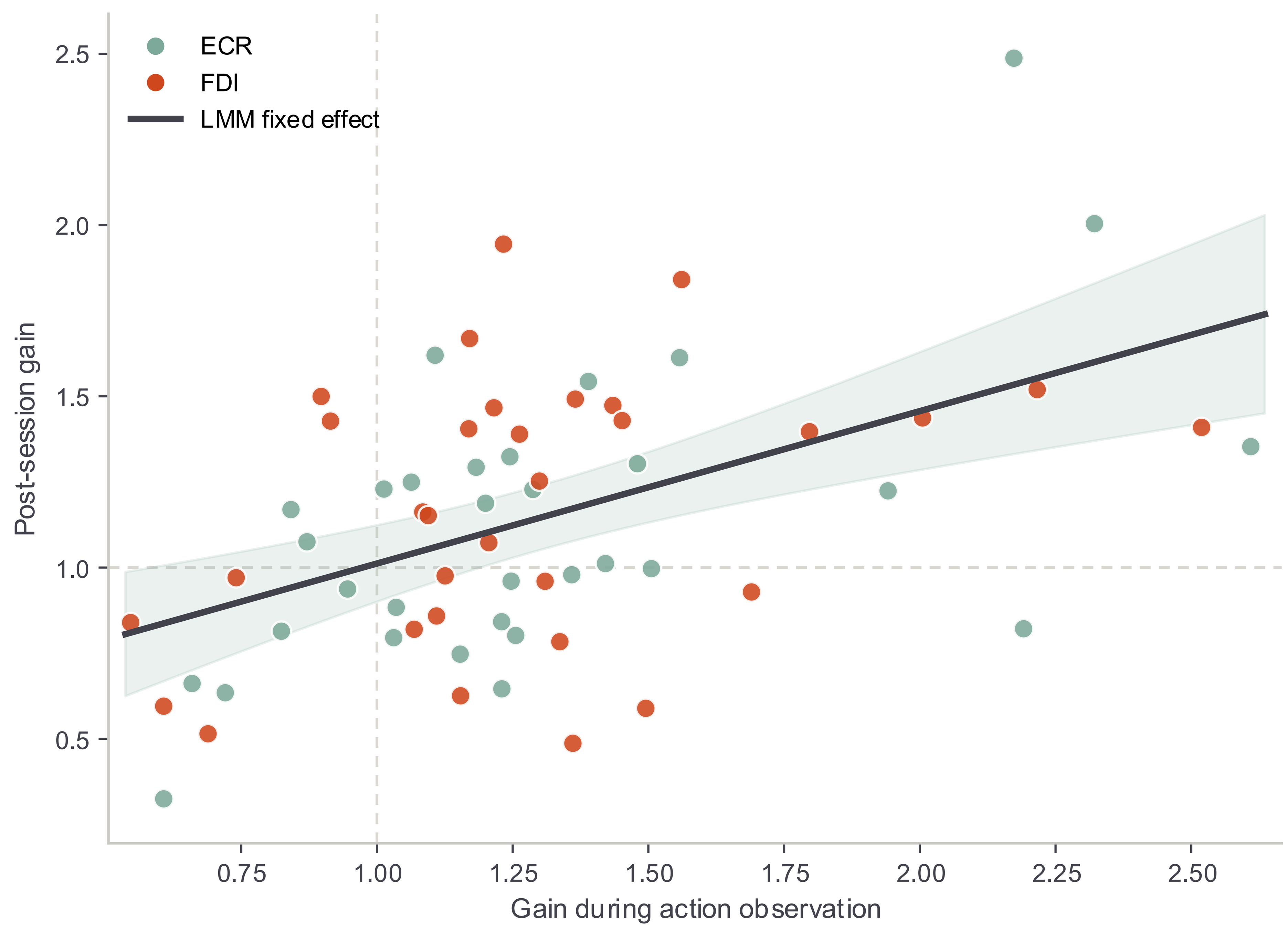
ONLINE FACILITATION → AFTER-EFFECT

Both indices: MEP gain relative to baseline rest

Stronger online facilitation predicts stronger post-session corticospinal excitability

β = 0.445 | **$p < .001$** | **$R^2_m = .247$**

Participants showing stronger facilitation during AO also showed stronger post-session excitability.

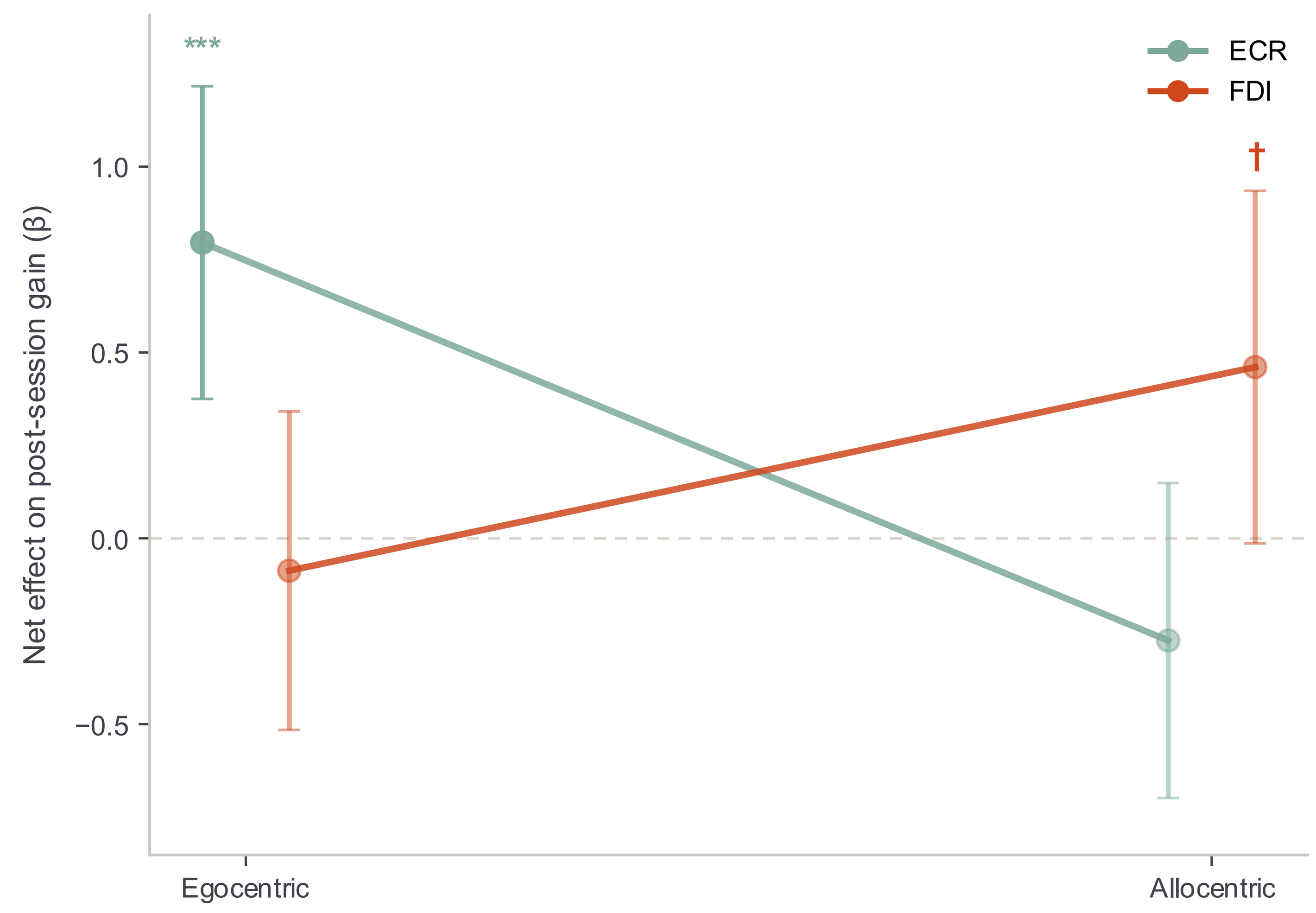


Action observation reliably facilitates Corticospinal Excitability

Facilitation was consistent across the four viewing conditions.

≈1.33× baseline
mean corticospinal excitability during AO
No differences across viewing conditions
ECR $p = .322$ · FDI $p = .409$

The online–after relationship shows perspective–muscle specificity



Perspective × Muscle $p = .014$

Egocentric → ECR β = .796 $p < .001$

Allocentric → FDI β = .461 $p = .057$ (trend)

Motor resonance does not end when observation stops.

Stronger online AO facilitation predicts stronger post-session CSE, supporting temporal continuity between online motor resonance and short-term after-effects.

AOT IMPLICATION
Perspective–muscle specificity may help guide protocol design.

NEXT STEP
Sham-TMS control to disentangle AO-driven from TMS-related after-effects.

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Poster & Abstract